

Active Directory Deployment & Automation Lab- Windows Server Environment

Jorge Antonio Nava

February 2026

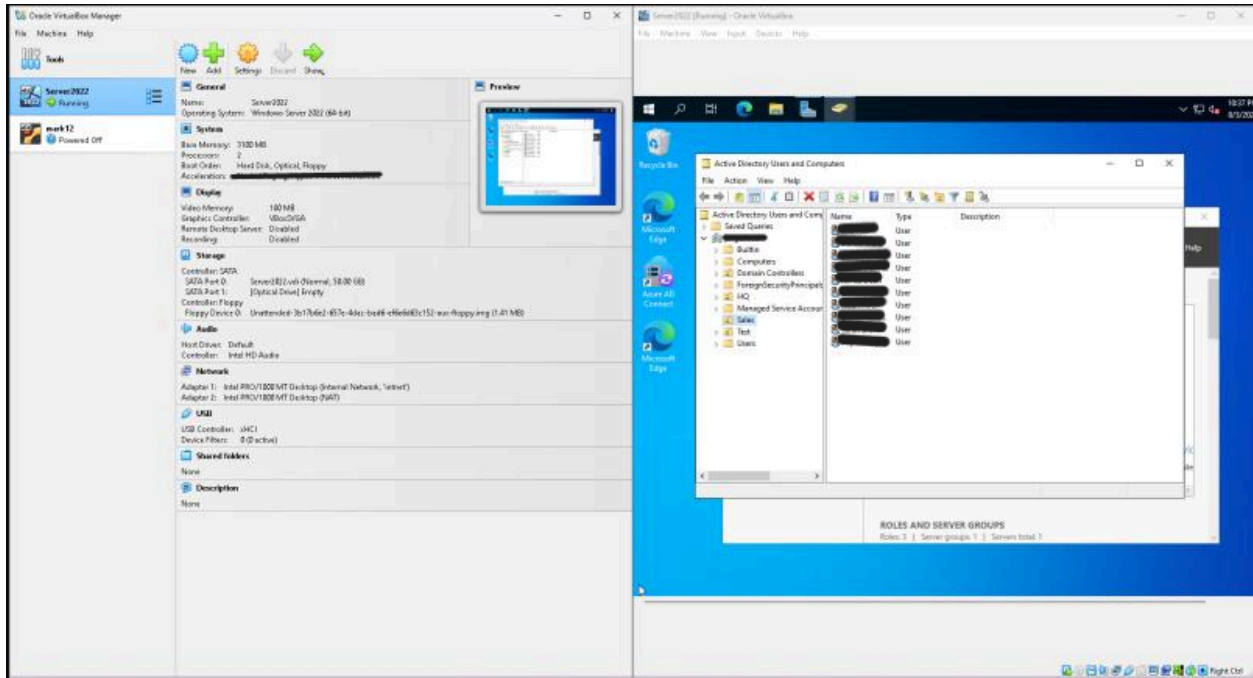
Tools & Environment: Oracle Virtual Machines (2x: Windows Server and Client Windows OS)

Overview: Built a 2-VM Active Directory environment simulating enterprise onboarding: domain controller and client, OU/group structuring, GPO security baselines, and role-based file permissions. Automated bulk user provisioning via PowerShell (CSV import), cutting manual ADUC entry from 10 individual account creations to one script execution. Diagnosed and resolved an NTFS/share permission conflict that caused incorrect folder access, and a CSV path misconfiguration during script testing.

What This Demonstrates:

- Windows Server / AD DS deployment
- PowerShell scripting for bulk operations
- GPO security policy enforcement (password/lockout policies)
- NTFS vs. share permission troubleshooting
- Root-cause diagnosis of access issues

Windows Server 2022 Virtual Machine Deployment



Two Virtual Machines required, one for the server and the other VM for the client

Oracle VM Network Adapter Configurations

Server

- Adapter 1 = Internal Network (named LabNet) → static IP 192.168.10.10
- Adapter 2 = NAT (for Server internet/updates)

Server Static IP & DNS Configuration

- IP: 192.168.10.10 / Mask: 255.255.255.0 / DNS: 192.168.10.10

Client

- Adapter 1 = Internal Network (named LabNet) → static IP 192.168.10.11
- Adapter 2 = NAT (for Client internet)

PowerShell Bulk Scripting - Troubleshooting

Issue: 7+ end users reported to be in the wrong department or cannot access/ find shared resources.

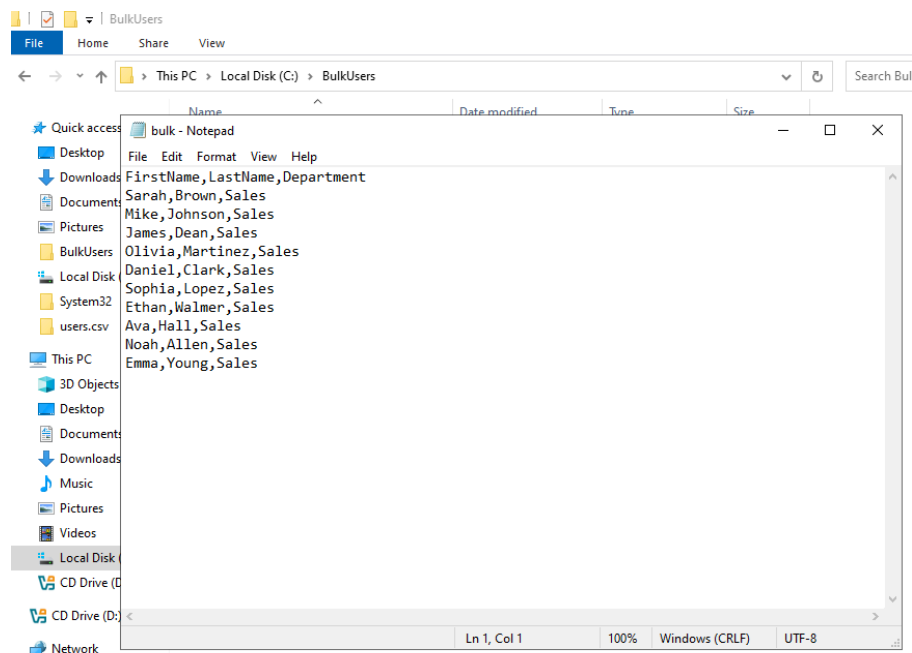
Diagnosis: Current users have access to the wrong department due to incorrect OU assignment/ faulty bulk scripting. Symptoms can be seen on ADUC, specific users are in the wrong OU/ group..

Fix: Created a new Bulk Script to create new users and add them to the existing organizational units/ groups. Used PowerShell commands to remove users from the wrong OU first, then updating AD infrastructure using the PowerShell script provided.

Bulk Script Used to Create New Users and Add Them to Existing OUs and Group

```
Administrator: Windows PowerShell
>>> -Name "$($user.FirstName) $($user.LastName)" `
>>> -GivenName $user.FirstName `
>>> -Surname $user.LastName `
>>> -SamAccountName $username `
>>> -UserPrincipalName "$username@dnsRoot" `
>>> -Department "Sales" `
>>> -AccountPassword $password `
>>> -Enabled $true `
>>> -ChangePasswordAtLogon $true `
>>> -Path $ouPath `
>>>
>>> Add-ADGroupMember -Identity $groupName -Members $username
>>>
>>> Write-Host "Created and added to Sales_Group: $username" -ForegroundColor Green
>>> }
>>> else {
>>>     Write-Host "User already exists: $username" -ForegroundColor Yellow
>>> }
>>> }
>>>
>>> created and added to Sales_Group: sarah.brown
>>> created and added to Sales_Group: mike.johnson
>>> created and added to Sales_Group: james.dean
>>> created and added to Sales_Group: olivia.martinez
>>> created and added to Sales_Group: daniel.clark
>>> created and added to Sales_Group: sophia.lopez
>>> created and added to Sales_Group: ethan.walmer
>>> created and added to Sales_Group: ava.hall
>>> created and added to Sales_Group: noah.allen
>>> created and added to Sales_Group: emma.young
PS C:\Users\Administrator>
```

Created New List of Users, Updating New CSV File via Notepad:



PowerShell Script Sample Provided

```
Import-Module ActiveDirectory

# Get domain information automatically

$domain = Get-ADDomain

$baseDN = $domain.DistinguishedName

$dnsRoot = $domain.DNSRoot


# Define existing OU and Group

$ouPath = "OU=Sales,$baseDN"

$groupName = "Sales_Group"


# Default password

$password = ConvertTo-SecureString "Password101!" -AsPlainText -Force


# Import CSV

$users = Import-Csv "C:\BulkUsers\bulk.csv"


foreach ($user in $users) {

    $username = ($user.FirstName + "." + $user.LastName).ToLower()

    # Check if user already exists

    if (-not (Get-ADUser -Filter "SamAccountName -eq '$username'" -ErrorAction SilentlyContinue)) {

        New-ADUser `

            -Name "$($user.FirstName) $($user.LastName)" `
```

```

-GivenName $user.FirstName `
-Surname $user.LastName `
-SamAccountName $username `
-UserPrincipalName "$username@$dnsRoot" `
-Department "Sales" `
-AccountPassword $password `
-Enabled $true `
-ChangePasswordAtLogon $true `
-Path $ouPath

```

```
Add-ADGroupMember -Identity $groupName -Members $username
```

```
Write-Host "Created and added to Sales_Group: $username" -ForegroundColor Green
```

```
}
```

```
else {
```

```
Write-Host "User already exists: $username" -ForegroundColor Yellow
```

```
}
```

```
}
```

Commands for Deleting Accidentally-Created OUs:

- **Removes Protected OU from being deleted:**

```

Set-ADOrganizationalUnit `
-Identity "OU=NameOfOU,DC=Fabrikam,DC=com" `
-ProtectedFromAccidentalDeletion $false

```

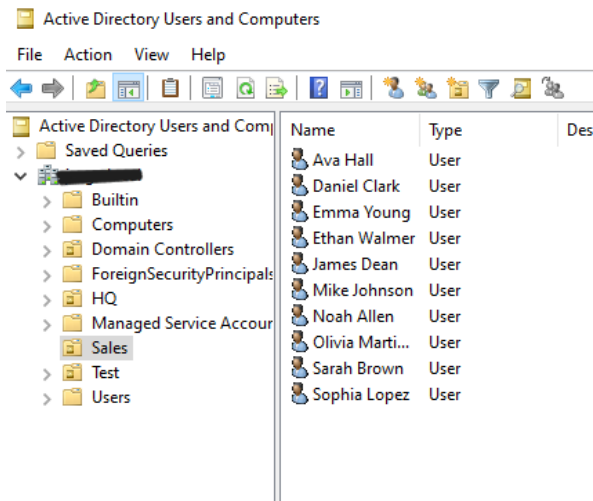
- **Completely Removes Specific OU:**

```

Remove-ADOrganizationalUnit `
-Identity "OU=NameOfOU,DC=Fabrikam,DC=com" `
-Recursive `
-Confirm:$false

```

Result: PowerShell script assigned users to their appropriate OU departments



Takeaway: Always check the original PowerShell bulk script to prevent users from being assigned to the wrong group/ shared resources. If an issue has occurred, remove users from OU and add them to the correct group using updated CSV file and PowerShell bulk script. Once the issue has been resolved, check ADUC to confirm the user is in the correct OU/ group. Issue resolved within 25 minutes.

Skills Learned:

- The Bulk Script below allowed me to add multiple users simultaneously into an Organizational Unit via 'Sales_Group'.
- CSV files filled with First Name, Last Name, and Department can be a reliable tool when adding multiple people without having to assign all of them through ADUC.
- It is more feasible to create a bulk script using the Csv file you created through File Explorer.
- Created a Share folder named "Scripts" or "BulkUsers" in the C: local drive (Example of Share Folder Path: "C:\BulkUsers\bulk.csv"). On a separate notepad, type or copy/paste the user names and their appropriate OU department. Ensure to name the notepad/ csv file (bulk.csv).
- Added file path into bulk script. I made sure the PowerShell script was tailored in a manner that allowed several users (10 max) to be added into the "Sales" Department.

Below demonstrates the success of the bulking script and an overview of the commands used.

Reset Passwords for Clients via ADUC & Group Policy Management - Password Policies

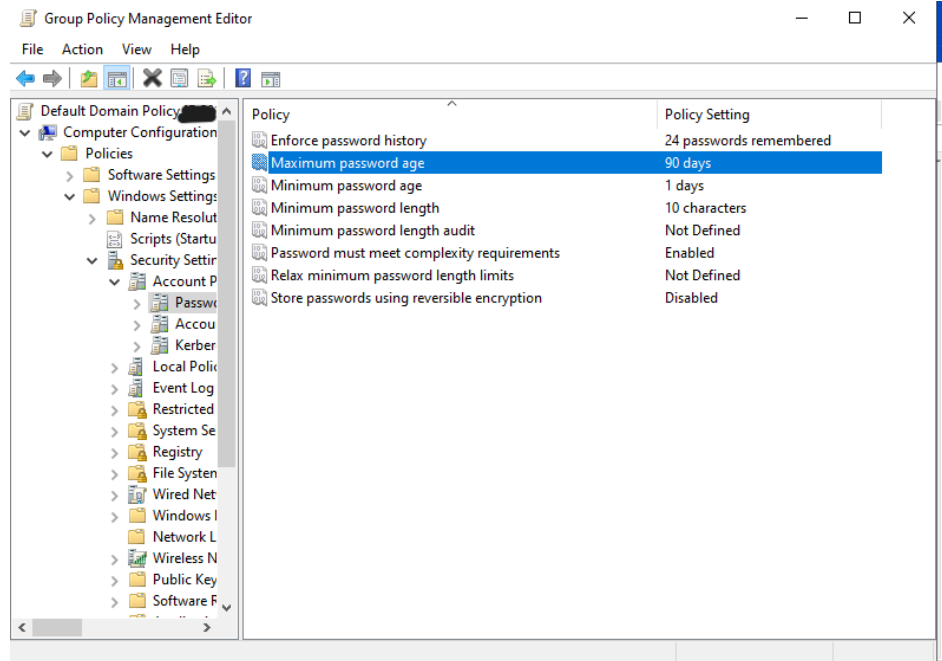
Issue: User James Smith cannot login using password, claims they have used the same password since they onboarded. Password policies are inconsistent, 6+ users get locked out frequently.

Diagnosis: User's password credentials have expired and password policies are not compliant with company standards.

Fix: Reset password for new/ existing users. On ADUC, I clicked OU of the user (HQ- OU) and found the user, right-clicked the user's name and clicked **Reset Password**. To prevent this issue from occurring persistently, I updated the Group Policy Management (GPO) password policies and account lockout policies.

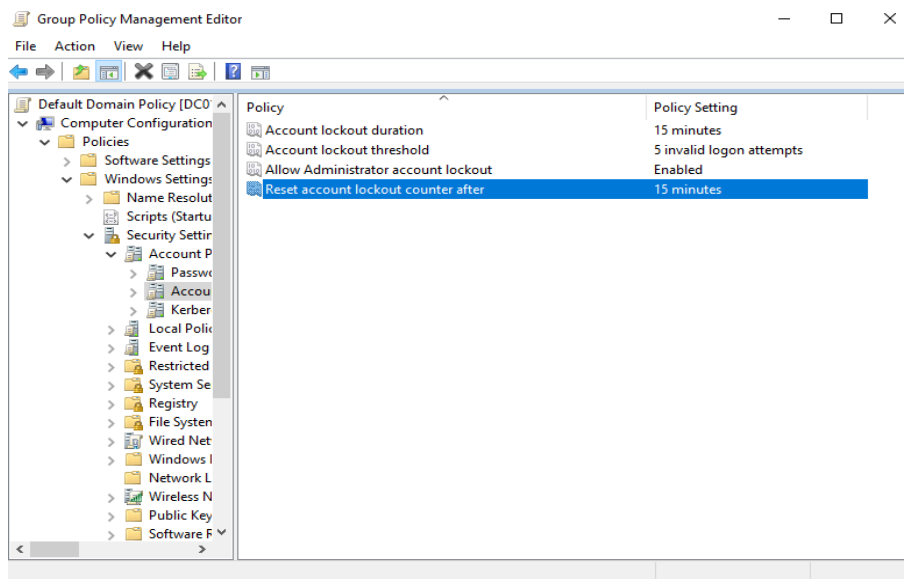
GPO Password Policies Min/ Max

- Minimum Password Length: 10
- Password must meet complexity requirements: Enabled
- Maximum password age: 90 days



Configure Account Lockout: Account Policies

- Account lockout threshold: 5
- Lockout duration: 15 minutes
- Reset counter after: 15 minutes



Verify New GPO Password Policies- Test the GPO on the Client's Workstation

- After GPO and password policies were implemented, ensure Client machine's new policies are up to date through the command prompt 'gpresult /r' or 'gpupdate /force' to force policy.

- Log in to User: Jane Smith's (jsmith) account, force policy update: Open Command Prompt→ type 'gpupdate /force'→ Reboot if needed
- Verify Applied Policies: 'gpresult /r'

```
C:\Users\jsmith>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

```
Created on 2/10/2026 at 11:49:50 AM

GPO data for \Administrator on : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:             10.0.17134.1
Site Name:              Default-First-Site-Name
Roaming Profile:        N/A
Local Profile:          C:\Users\Administrator
Connected over a slow link?: No

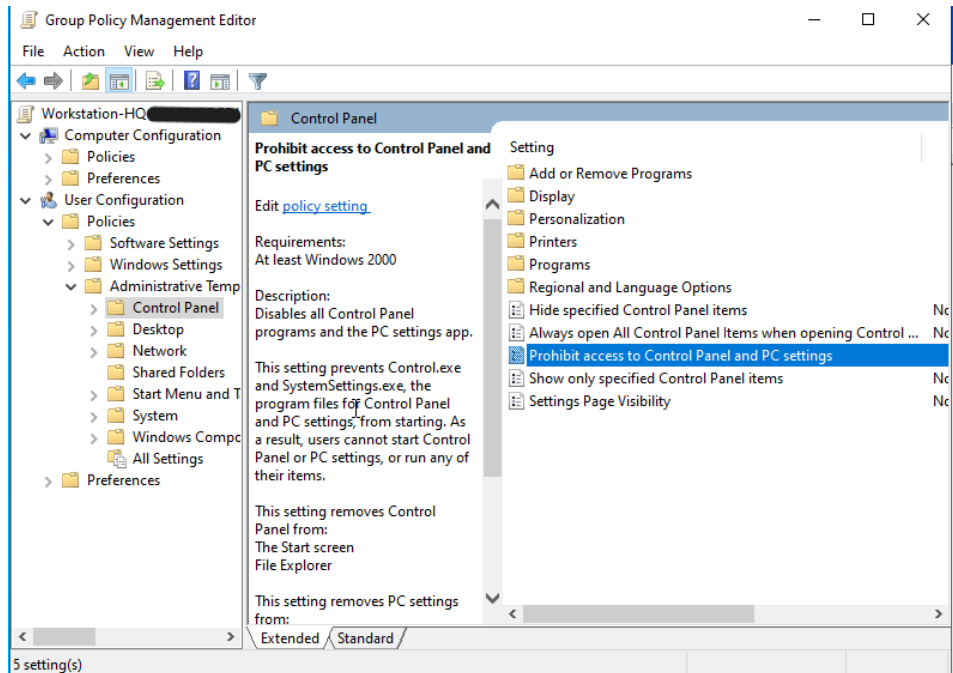
COMPUTER SETTINGS
-----
CN=,CN=Computers,DC=,DC=
Last time Group Policy was applied:  at
Group Policy was applied from:      local
Group Policy slow link threshold:   500 kbps
Domain Name:                       2026
Domain Type:                       Windows 2008 or later

Applied Group Policy Objects
-----
Default Domain Policy
```

- If error persists after running gpresult commands (such as lack of network connectivity), ensure Client machine has a default gateway that matches the IPv4 address of Domain Controller (Windows Server machine).
- Both machines must be able to ping each other to enforce new GPO policies and validate these policies through gpresult commands.

Securing Shared Resources & Reinforcing Least Privileges

For Individual User, Screen Lock After 10 Minutes



User Configuration→ Policies→ Administrative Templates→ Control Panel→ Personalization

- Enable screen saver: Enabled
- Screen saver timeout: 600
- Password protect the screen saver: Enabled

Disable Local Admin Access

- Navigated to: Computer Configuration→ Policies→ Windows Settings→ Security Settings→ Restricted Groups
- Right-click→ **Add Group**→ Add: Administrators
- In Members of this group, add:
 - Domain IT Admins
- This will force only approved groups to be local admins

Takeaway:

- Ensure users can log in to their account and access their vital resources through NTFS and manage Group Policy permissions through the GPO Management tool or edit the workstation.

- Implement GPO's password policies, configure account lockout/ screen lock, disable local admin access, and verify password policies to ensure security and optimize account access for end users.
- Resolved password reset issue within 5 minutes and resolution time for adding GPO Password policies was under 10 minutes, services met under SLA standards

File Server & Permissions Troubleshooting

Issue: 8+ clients cannot access their own shared resources or share resources to external users in the same department. Some clients claimed to have access to the wrong shared folders from other departments, risking sensitive information being shared.

Diagnosis: Misconfigured NTFS permissions, such as naming the wrong folders (such as naming HR instead of HQ). It confused the client's file explorer over which folder they should gain access to. Sometimes misconfigurations may lead users to gain access to the wrong shared folders.

To remediate this common issue:

-Did you assign any Groups?

- Having Organizational Units alone cannot find the Object name for the type of group you want to share.
- Make sure group names are correctly labeled

-Check if network path are valid:

- Open Command Prompt, type: net share
- Confirm which folders are actually shared

-Check NTFS Permissions:

- Right click-folder, then click Properties, then click Advanced.
- Remove broad groups, such as Users, Domain Users, Authenticated Users, Everyone. Only designated security groups such as SYSTEM/ Administrator/ CREATOR OWNER must remain.
- Make sure permission type is "Modify" for the group, Apply to: This folder, subfolders, and files.

-Check Share Permissions:

- Right-click folder, go to properties→Sharing→Advanced Sharing→ Permissions
- Check if only designated groups and Group type (HQ_Group and/ or Test_Group) are listed. Full control is valid for Groups.

-Confirm User Membership

- Open Active Directory to find the user, then click on Member Of tab
- Make sure the user is a member of their own group. Domain Users is fine as long as it does not have special NTFS permissions

-Nested Folder Warning

- Nested folder warnings with network paths such as \\DC01\HQ\Test (different Groups sharing one path = wrong configuration) can cause confusion.
- It may occur due to misconfiguration or common SMB behavior. Can be fixed by creating Share Test folder as its own share to separate paths (ex: \\DC01\Test)

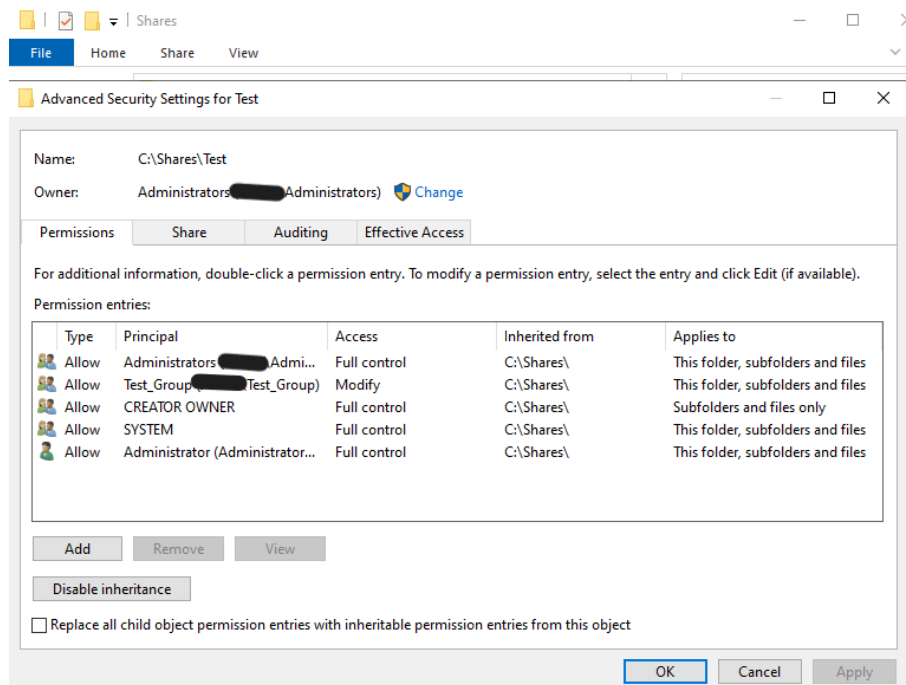
Fix: NTFS Configuration

On the Domain Controller (Windows Server VM), create C:/ Share folder, then create:

- C:\Shares\HQ
- C:\Shares\Test

Configure NTFS Permissions- Security Tab

- Under the Security tab, I clicked Advanced and disable inheritance. Made sure to click “Convert inherited permissions”
- Remove “Users” or “Everyone” and Add HQ_Group or Test_Group



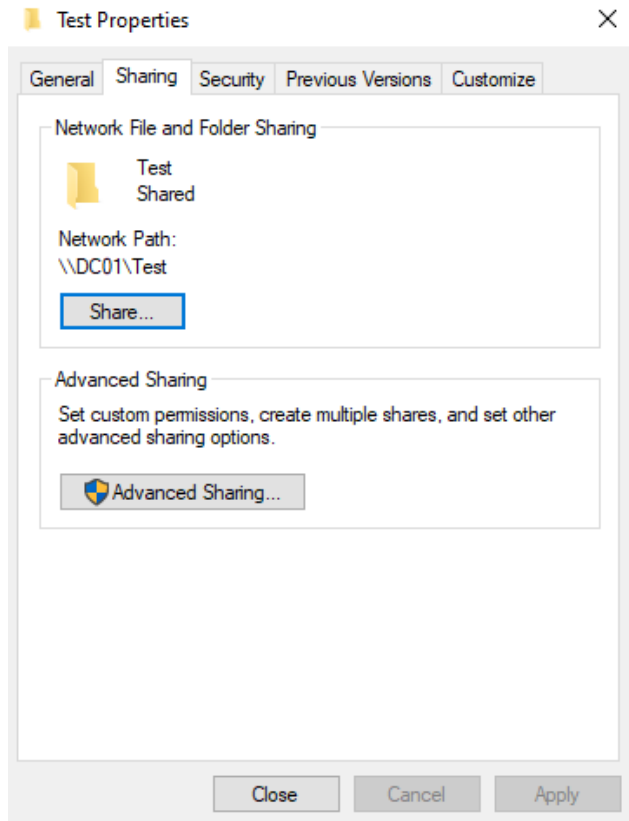
Shared the Folders- Sharing Tab

- Share the folders on the Sharing tab

- Advanced Sharing, then clicked Check Share this folder, then add share name
- Click Permissions, remove Everyone and add HQ_Group (for HQ folder) and Test_Group (for Test folder)

Network paths

- For HQ folder: \\DC01\HQ
- For Test folder: \\DC01\Test

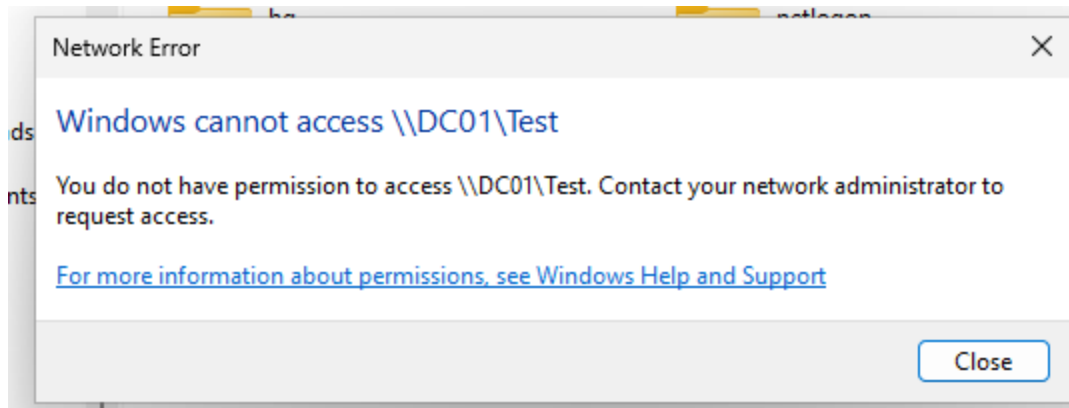


Test access

- Access HQ or Test user account and see if folder share works. Ensure users can only access shared folders they have permission to.

Verify NTFS Permissions:

- Access Test user account and see if Test share works
- Test users will not have access to the HQ folder.



- NTFS Permission Access across Multiple Users: HQ user cannot access Test shared folder (user only has permission to access HQ folder)

Takeaway: Ensure that users have access to shared resources related to their department/ work role. It is important to uphold least privileges and to correct any NTFS misconfigurations to prevent information from being shared to the wrong department. Users must only have access to shared resources that are required to do their job. Resolution time in 20 minutes, including changing file path and updating NTFS permissions for 10+ users.

Project Notes/ Observations:

The Active Directory Deployment & Automation Lab was a success involving assigning new users to their correct departments through Active Directory Users & Computers tool and creating a PowerShell bulking script for assigning end users into existing OUs and groups. Certain measurements were taken before implementing the bulking script, such as setting the correct Csv log path through NTFS permissions, ensuring RBAC privileges were up to date.

In case an OU was accidentally created and it requires an immediate removal, I ensure to type a command that removes its protection from accidental deletion, then fully delete the OU or built-in container that I created by accident.

When running the bulking script, I made sure no errors, such as “Access to ___path is denied”, were present. If so, see if you’re running PowerShell as the Administrator or if NTFS permissions are correctly configured. In my scenario, I mistyped the name of the csv file and saved it into the wrong file path. From learning how to troubleshoot this issue, I had to retrace my steps and find which procedure I had made a mistake on (such as mistyping the file name and saving it to the wrong file path).

Optional Project Add-Ons/ Enhancements

- ☐ Create python text script to automate bulk users
- ☐ Help Desk Ticket Simulation
- ☐ PowerShell Automation Project -Automate user creation and group membership.
- ☐ Autopilot Deployment Project -True enterprise onboarding simulations.
- ☐ SIEM Integration with Microsoft Defender -Cloud native security monitoring.

Sources:

- [Remove-ADOrganizationalUnit \(ActiveDirectory\) | Microsoft Learn](#)
- [Add Users to Group with PowerShell - ALI TAJRAN](#)
- [How to Create Multiple Active Directory users in PowerShell](#)